

5	a	The rocket experiences equal and opposite gravitational forces exerted by Earth and Moon. Hence there is zero net force as the forces cancel out.	B1
	b	1. Moon does not have an atmosphere hence there is no need to supply energy to do work against air resistance. 2. The gain in gravitational potential energy from Moon to P is much lower than that from Earth to P hence less kinetic energy is required for the rocket to move from Moon to P.	B1 B1
	c	At P, $F_E = F_M$ $\frac{GM_E m}{d_E^2} = \frac{GM_M m}{d_M^2}$ $M_M = \frac{(5.97 \times 10^{24})(3.80 \times 10^7)^2}{(3.46 \times 10^8)^2}$ $= 7.20 \times 10^{22} \text{ kg}$	C1 A1
	di	The gravitational potential at a point is the work done per unit mass in bringing a point mass from infinity to that point.	B1
	dii	$\phi_P = -\frac{GM_E}{d_E} - \frac{GM_M}{d_M}$ $= -\left(6.67 \times 10^{-11}\right) \left(\frac{5.97 \times 10^{24}}{3.46 \times 10^8} + \frac{7.20 \times 10^{22}}{3.80 \times 10^7} \right)$ $= -1.277 \times 10^6 \text{ J kg}^{-1}$ KE = Loss in GPE $= (7.5) \left[0 - (-1.277 \times 10^6) \right]$ $= 9.58 \times 10^6 \text{ J}$	C1 C1 A1

6	ai	Total resistance of the 4 parallel resistors = $(1/2R + 1/2R)^{-1} = R$	R1
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